

Q8 acceleration, $a = a_0 \sin \omega t$

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displacement, $x = -x_0 \sin \omega t$

$$= -x_0 \sin \left[\left(\frac{2\pi}{T} \right) \left(\frac{T}{3} \right) \right]$$

$$= -x_0 \sin \left(\frac{2\pi}{3} \right)$$

$$= -0.87x_0$$

Answer: A

Q8 A and B are correct as they are valid assumptions in the simple kinetic theory of gases